Wasti Murad

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SUMMARY

Recent MSc Data Science graduate specializing in end-to-end machine learning systems, predictive modeling, and big data applications. Experienced in model optimization and interpretable AI solutions, with a focus on automated deployment and experimentation frameworks. Strong background in statistical validation, cloud-based workflows, and data-driven storytelling, translating complex analytical results into actionable insights. A methodical problem-solver who applies the same critical thinking and strategic reasoning to data challenges as to planning a few moves ahead in chess.

EDUCATION

Maastricht University, Limburg, Maastricht, Netherlands — Sep 2023 – Aug 2025
MSc Data Science for Decision Making — GPA 3.7/4.0
Key courses: Computer Vision, Deep Learning, Data Mining, Computational Statistics

Jahangirnagar University, Savar, Dhaka, Bangladesh — Mar 2017 – Sep 2022
BSc Physics — GPA 3.2/4.0
Key courses: Quantum Mechanics, Nuclear Physics, Solid State Physics, Mathematical Physics

SKILLS

Programming and Tools: Python, SQL, Excel, Git, GitHub, Tableau, Power BI
Machine Learning and AI: LangChain, RAG, LLMs, PyTorch, TensorFlow, Scikit-Learn, Computer Vision, Transfer Learning, Time Series, Predictive Modeling
Data Science Libraries: Pandas, NumPy, SciPy, Statsmodels, OpenCV
Big Data Processing: Apache Spark, PySpark, Spark Optimization (Partitioning, Broadcast Joins, Caching, AQE)
Data Engineering: Databricks, Delta Lake, Delta Live Tables, Medallion Architecture, Unity Catalog
Cloud and MLOps: Azure Data Factory, AWS, SageMaker, MLflow, Hydra, Docker, GitHub Actions, CI/CD
Experimentation and Analytics: A/B Testing, Statistical Testing, Power Analysis, Sequential Testing
Languages: Bengali (Native), English (C1), Dutch (A1)
Professional: Communication, Teamwork, Project Management, Reproducibility, Time Management

EXPERIENCE

Master's Thesis – Artificial Intelligence Guided Adaptive Radiotherapy Using Counterfactual Explanations in Latent Space

Maastricht University and Maastro (Clinical Data Science) — *Maastricht, Netherlands* — Feb 2025 – Aug 2025

- Built a 3D Variational Autoencoder with an attention mechanism that generates anatomy-preserving counterfactuals, solving the lack of interpretable explanations in radiotherapy AI workflows and improving reconstruction quality by 4%
- Restructured the latent space to separate tumor from organ variations, solving the entanglement problem that obscured what drove model behavior and enabling clinician-validated traversals ready for medical review
- Established an MLflow-based experimentation workflow that solved inconsistent run tracking across team members, delivering a documented, reproducible pipeline for collaborative research development

Technologies: Python, PyTorch, CUDA, MONAI, nibabel, MLflow

Machine Learning Intern – Model Generalization and Data Pipelines

Maastricht University Medical Center — *Maastricht, Netherlands* — Sep 2024 – Feb 2025

- Enhanced preprocessing and class-balancing pipelines to solve training instability on highly imbalanced medical datasets, improving model reliability and generalization on unseen patient data
- Developed a Hydra and MLflow experimentation framework that replaced fragmented configuration management and manual tracking, delivering reproducible experiments and smoother team collaboration
- Produced clinician-friendly Explainable AI summaries that closed the gap between technical model outputs and clinical interpretation, turning raw results into stakeholder-ready insights that supported feedback-driven model improvement

Technologies: Python, Pandas, AutoML, Hydra, Git, GitHub

RESEARCH EXPERIENCE

Personal Wildlife Monitoring (Group Research Project)

Maastricht University — *Maastricht, Netherlands* — Aug 2023 – Jul 2024

- Architected an edge-to-cloud pipeline using motion and audio triggers that eliminated manual wildlife-stream screening with automated detection, cutting the time to identify key moments from hours to minutes
- Integrated a real-time object detector, Transformer-based models, and acoustic features into a multimodal detection pipeline deployed behind a containerized FastAPI backend, enabling continuous real-time monitoring instead of post-hoc review

Technologies: Python, OpenCV, Librosa, Transformers, Docker

TECHNICAL PROJECTS

Code Repository

Causal Inference and Uplift Modeling for Marketing Campaign Optimization

- Estimated individual treatment effects on observational marketing data, correcting non-random-exposure bias to reveal each user's true incremental lift rather than correlation-inflated estimates
- Surfaced a 25–30% high-uplift segment (high prior-spend, Outdoor/Fashion) and built a Target, Don't-target, or Exclude framework, informing data-driven campaign recommendations that moved beyond broad targeting into causal, profit-optimized allocation

Technologies: Python, Pandas, Scikit-learn, Statsmodels, CausalML, Propensity Scoring, T-Learner, Uplift Random Forest

LLM-Powered E-Commerce Analytics Assistant with RAG and Evaluation Framework

- Built a query router dispatching each question to a Pandas engine, RAG retriever, or hybrid pipeline, solving the hallucinated-numbers problem on analytical queries and keeping answers accurate across data types
- Combined ChromaDB MMR retrieval with a Pandas engine and context-strict GPT-4o prompts, answering mixed sales-and-review questions that neither SQL nor plain RAG can handle alone
- Added an LLM-as-judge evaluator scoring faithfulness, relevance, and hallucination, with a YAML config for code-free model swaps, closing the quality blind spot in most LLM apps and enabling faster model comparisons

Technologies: Python, LangChain, OpenAI (GPT-4o), ChromaDB, MMR, Pandas, Streamlit, YAML

Apple Retail Sales and Warranty Analytics

- Uncovered revenue concentration and seasonal demand patterns across multi-year Apple retail data using advanced SQL, showing that the top 20% of stores generated 40% of revenue and guiding store-prioritization decisions
- Quantified product reliability through 180-day warranty-claim cohorts, revealing that mid-tier products had 35% fewer claims than premium ones, and delivered Tableau dashboards for ongoing regional trend tracking

Technologies: SQL, Window Functions, CTEs, Joins, Tableau

Databricks Delta Live Tables: Streaming ELT Medallion Pipeline

- Engineered a streaming ELT pipeline (Bronze/Silver/Gold layers, ACID transactions, change data capture, declarative data-quality expectations) that replaced brittle batch workflows with a reliable real-time architecture that enforces schema and quality at every stage
- Designed PySpark transformation logic for joins, derived metrics, and type casting, and versioned the full pipeline through Git and GitHub inside Databricks, producing reproducible, peer-reviewable data engineering work

Technologies: Databricks, PySpark, Delta Live Tables, Delta Lake, Medallion Architecture, SQL

End-to-End Machine Learning Pipeline with Automated Training and Deployment

- Built a GitHub Actions and Docker CI/CD pipeline that automates training, testing, and deployment on every commit, replacing manual release steps with a reproducible, production-ready workflow
- Deployed a real-time inference API with performance monitoring, providing continuous visibility into model behavior and enabling faster detection of issues than ad-hoc checking

Technologies: Docker, scikit-learn, GitHub Actions, CI/CD, MLOps, pytest, FastAPI

CERTIFICATIONS

Machine Learning Specialization – Stanford University (Coursera)
Complete Python Developer – Zero to Mastery Academy
Tableau 2020 A-Z: Hands-On Tableau Training – Udemy

AWARDS

International University Physics Competition 2019, 2020

- Bronze Medal (2020): Problem A – Protecting Travelers to Mars
- Silver Medal (2019): Problem A – Ion Thrusters to Saturn